

# AHMED HAMDY ABDELAZIZ

Data Scientist | Machine Learning Engineer

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[LinkedIn Profile](#) [Web Portfolio](#)

## SUMMARY

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Data Scientist & Machine Learning Engineer with hands-on experience building and deploying end-to-end ML solutions — from data preprocessing and feature engineering to production-ready web applications. Skilled across the full pipeline: EDA, model development (TensorFlow, Keras, Scikit-Learn), and deployment via REST APIs. Currently expanding into MLOps practices (model versioning, CI/CD for ML, experiment tracking with MLflow) and cloud deployment on AWS/GCP. Passionate about turning data into real-world, accessible products.

## PROJECTS & APPLIED EXPERIENCE

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### Hieroglyphs-AI — Full-Stack AI Translator

[GitHub Repository](#) [Live Demo](#)

- Designed and deployed a full-stack web app (Next.js + FastAPI) that classifies Egyptian hieroglyphs from photos in real time, returning the Gardiner code, name, and confidence score
- Fine-tuned an InceptionV3 CNN and built a REST API with 5 endpoints (predict, symbols, categories, health) serving a database of 1,000+ Gardiner-classified symbols
- Built a rich, responsive frontend with an interactive translator, gallery, timeline, and 3D artifact viewer
- Containerized the backend with Docker and deployed across Vercel (frontend) and Render (backend) with CI/CD
- Tech Stack: Python, TensorFlow/Keras, FastAPI, Next.js, React, Tailwind CSS, Docker, Vercel, Render

### SpaceX Falcon 9 Landing Prediction | *Completed as part of IBM's Applied Data Science Capstone (Coursera)*

[GitHub Repository](#)

- Built an end-to-end data science pipeline to predict the success of Falcon 9 first-stage landings, from data acquisition to modeling and evaluation
- Collected and cleaned launch data via Web Scraping and the SpaceX REST API; performed data wrangling and exploratory data analysis (EDA)
- Trained and compared multiple classification models (Logistic Regression, SVM, Decision Tree, KNN), selecting the best performer based on accuracy
- Tech Stack: Python, Pandas, Scikit-learn, SQL, Folium, Plotly Dash

## WORK EXPERIENCE

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### Machine Learning Trainee | *Digital Egypt Pioneers Initiative (DEPI) - Microsoft Track* July 2026 – Present

- Designed and evaluated diverse machine learning algorithms, focusing on model optimization, hyperparameter tuning, and cross-validation techniques to achieve robust predictive performance
- Collaborated on intensive applied data science projects, utilizing modern ML frameworks to solve complex analytical tasks and streamline model deployment workflows

## EDUCATION

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SadatAcademy For Management Sciences — Computer Science, Level 3 | Student

GPA: 3.64 / 4

## SKILLS

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- Programming Languages: Python, C++, JavaScript, SQL
- ML/DL Frameworks: TensorFlow, Keras, PyTorch, Scikit-learn
- Data Analysis: Pandas, NumPy, Matplotlib, Plotly
- Tools & Platforms: MLflow, Anaconda, GitHub, Power BI, Streamlit
- Web: HTML, CSS

## CERTIFICATIONS

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- IBM Data Science Professional Certificate — IBM / Coursera (2025)
- [View Certificate](#)
- Applied Data Science Capstone — IBM / Coursera (Aug 2025)
- [Verify Certificate](#)
- Python for Data Science & AI — IBM / Coursera
- Data Analysis with Python — IBM / Coursera
- Machine Learning with Python — IBM / Coursera

## ADDITIONAL INFORMATION

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Languages: Arabic (Native), English (Intermediate)